

TrailMate: Towards safer hiking through social presence

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(a) The idle state of a hikers' smart bracelet



(b) The connected partner bracelet

Figure 1: A system of two smart bracelets for safe hiking called 'TrailMate'

ABSTRACT

Marginalized groups are often underrepresented in solo-hiking. This is partly due to fear of harassment, accidental injury and getting lost. To address this, a two-way bracelet system was conceptualized. A low-fidelity prototype was made utilizing the wizard-of-oz technique that through pilot evaluations gathered valuable data that defined the next iteration phase. The functionality was streamlined, focusing on unambiguous feedback and ease-of-use in stressful situations. The system consists of two identical bracelets: one for the hiker, one for a trusted friend, each equipped with home- and hike-mode. The main functions are; GPS tracking, self-indicating emergency signaling, safe signaling through regular check-ins and check-ins from partner bracelets, status indication with color, vibration and sound. Final evaluations revealed that the bracelet enhances the sense of safety and companionship through the presence of social interaction. However, several concerns were voiced. Some were scared the device might aggravate a potential attacker and others reflected on that the design needed to be more delicate. Further iterations are needed in order to ensure the inclusion of the entire target group. Other potential areas of improvement are technical advancements such as monitoring vital signs. Overall the product was met with appreciation, being described as effective and comforting.

CCS CONCEPTS

• Human-centered computing → Interaction design.

KEYWORDS

Prototyping, Interaction, Design, Wearable, Hiking, Safety

1 INTRODUCTION

Hiking has often been described as a panacea for health, and yet it isn't for everybody [11]. Only a small group of people are able to enjoy hiking as a leisure activity. Namely the able bodied, white, cis, male body. Whilst other groups such as women, queer, people of color and so forth are underrepresented. Especially in solo-hiking. This is partly due to harassment or fear of harassment [11], and partly because of fear of accidental injury or getting lost [3]. To address this issue a research question was formulated.

How can a wearable device favor solo hikers, fostering a sense of companionship and improving safety and emotional well-being during their hikes?

The aim of this project was, through prototyping and user-evaluations to come up with a product that would enhance the sense of safety for individuals, particularly women and other minority groups, when engaging in hiking. The idea aimed to foster comfort through subtle, ongoing communication between the solo hiker (see Figure 1a) and their partner at home (see Figure 1b), and to offer control through being able to send distress signals in dangerous situations with direct and clear feedback.

Prototyping played an important role in communicating with users by allowing the design and functionality of the TrailMate bracelet to be tested and refined based on user feedback. By creating an initial low-fidelity prototype, together with a written scenario, insight was gathered into how users interacted with the bracelet's key features, such as check-ins, alerts, and distress signals. An iterative process made the device more intuitive and responsive to

real-world scenarios. Through adjustments to features like response times, color signals, and vibration alerts, a working high-fidelity prototype was created. Further user tests revealed in depth insights to further improvements and future work.

2 RELATED WORK

Since solo hiking is a widespread problem for women and minority groups [11], several services/products have been made trying to combat this issue. Sexual harassment is a big problem in India, and a security system consisting of a wearable band and an app was developed that would both injure the attacker and guide the victim to safety [7]. Another solution to the issue was using IoT technology to monitor the location of women and send distress SMS text messages to authorities [6].

Considering ethics with a violent solution like the taser, the wearer might not feel morally okay with using a product like that. At the same time the weapon might be used by the attacker just as well by the victim. Another thing to consider is the socio-economic aspect. Hiking is generally an affordable sport, not requiring expensive equipment or special facilities. Owning a smartphone and having access to mobile data can be seen as a luxury, and it's not certain for all hikers. But then again, a product like this might also be a luxury even though it should not be, and one can argue that anyone who would buy a device for safe hiking would also afford a smartphone. An additional thing to reflect on is the integrity of the hiker. Would a direct contact with authorities make the hiker feel safe, or rather more scared to alert because of the repercussions if they are at fault. Moreover, Sicorello et al. found that social interactions, or even the presence of social interactions can lower stress levels in something called stress buffering [10].

During initial benchmarking, various solutions addressing outdoor safety were examined, before the focus later shifted toward enhancing the safety of women and minorities in solo hiking situations. Those solutions were an avalanche transceiver is a device used by skiers in groups to locate one another in case of an emergency [1]. A product with similar properties is the RECCO reflector, a small electronic chip that can be found by search and rescue teams through radar signals. The RECCO reflector can be found using helicopters, covering large areas fast [2].

3 INITIAL PROTOTYPING

3.1 Process

After mind mapping different sports who had, or lacked, social aspects, hiking was decided on. After continued dialog with the project criteria in mind the user group was narrowed down to women and minorities who engage in solo-hiking. The research question was formulated: How can a wearable device favor solo hikers, foster a sense of companionship and improve both safety and emotional well-being during hikes?

After that decision was made, the 6-3-5 brainwriting method was used to generate ideas [13], but instead of writing sketching was utilized. Each group member created three potential solutions to the problem statement, spending five minutes sketching each concept followed by a short presentation on the proposed solutions. Afterwards, the sketches were circulated among the group for improvements. Each member then cast five votes for their preferred

ideas. Using an ideation method as 6-3-5 brainwriting where each member contributes to each other's ideas and avoids idea critiquing creating a collective experience where no-one "owns" an idea [5]. This method was also good to focus on quantity over quality. However, using a decision matrix could have been beneficial in choosing the winning concept [8].

The two highest-rated concepts were a bracelet system, where a family member or friend signals the hiker through sound or vibration, requiring the hiker to press a button to confirm their safety, and a ring worn by both individuals that turns red in cases of distress or declining health (e.g., low pulse or oxygen), signaling potential danger to the other person. Since the two ideas were quite similar it was decided to combine the two into a bracelet system that had both of the functionalities of the original bracelet and ring ideas. Through discussions the bracelet was selected as the preferred wearable due to its comfort and practicality for hiking. Unlike a ring, which could cause discomfort or be difficult to use with gloves, the bracelet allows for better visibility of color indicators and easier detection of vibrations, enhancing both functionality and user experience. A further ideation session could have been helpful in order to properly explore the solution sphere and possibilities with combining the two concepts and what to explore further what type of wearable would have been best for that.

After that a low fidelity prototype using the sketches as reference was made. Two different bracelets were made of simple materials like tape, paper, string and pearls. One for the hiker (see Figure 2a) and one for the person who is not hiking, the latter utilized the wizard-of-oz method and pulled different colored paper through the bracelet (see Figures 2b, 2c). We used a variety of materials to enhance user engagement. This approach allows users to better understand and experience the product, facilitating a deeper connection. Additionally, the use of different materials aids in identifying potential challenges that may arise in the future, therefore informing our design process [12]. Creating quick low-fidelity prototypes made the different features of the bracelet more clear and an understanding of how they fit together was formed. Low fidelity prototyping was a useful tool to explore the solution sphere early on and continuously evaluate our work as it was a good medium for communication between designers and users [4].

Once the initial prototype had been created user feedback was required in order to improve and refine the design. Feedback was collected through conducting interviews, which execution and result is discussed in further detail in the next chapter. With the initial prototype in mind a simple video was also produced to demonstrate the prototype's functionality and illustrate the context of its use, proving invaluable in showcasing intended user interactions and situating the product within its real-world environment. This video effectively brought our user experience to life and clearly communicated the underlying design concepts.

3.2 Low-fidelity prototype

Several functions were discussed for the bracelet. To maintain attention towards the target user group and to offer a more focused functionality, it was decided to exclude the ideas of having crash detection and pulse and oxygen level monitoring since that seemed more relevant for extreme hikers, while our target group consisted

more of recreational hikers. Ultimately, we narrowed the design down to four essential functionalities:

- GPS tracking
- Self-indication tag (emergency signaling)
- Backup self-indication button (safe signaling)
- Status indication with color (good, check in, sos)

The hikers bracelet receives a scheduled “check-in” every set hours, e.g. every 2 hours. This check-in is answered by the push of a button. If the hiker does not answer, the other bracelets gets alerted and receive the hiker’s GPS coordinates. The same way the partner can issue the check-in at any moment, triggering the same alert with the hikers bracelet. The hikers bracelet also has a self-indication tag which can be pulled, immediately triggering the distress signal with the other bracelet (see Figure 2a). The self-indication tag is long so that it can stick out from underneath thick clothing and jackets. The non-hikers bracelet has a color status indication, where green means “ok” (see Figure 2b) and red means “emergency” (see Figure 2c). The bracelet also has a small screen that could display the coordinates of the hiker.



Figure 2: Prototypes of the hiker and non-hiker bracelets in various states

4 PILOT EVALUATIONS

4.1 Evaluation methods and execution

Following the development of the initial prototype, user feedback was gathered in order to evaluate the idea and features, without focusing too much on details. Two semi-structured interviews with open ended questions were held with participants aged 20-30. All of them had some hiking experience or participated in the activity regularly. One group interview with three women and one interview with a man. In these interviews, participants were presented with a scenario, after which they were asked open-ended questions regarding hiking alone in general and later regarding the product. What features they would prioritize, what would help them feel more safe and how they would like the bracelet to look and feel.

Our sample included a deliberate selection prioritizing women, given their relevance as our primary target group. However, we also included one male participant to gain additional insight from a broader perspective, despite men not being the main focus of our study. The scenario the participants were presented with was as follows:

Alex loves to go hiking, and does so every weekend. Today, they’re going on a hike in the mountain range outside their hometown. Alex is going alone since none of their friends could join this time, and felt a bit uneasy about it. So they put on their safety-bracelet and gave one to

their friend before setting off. It’s a nice day, and the hike is going well. After about two hours they stop for a coffee break but quickly notice someone walking towards them from a distance “They’re probably just going to pass, but still, you can never be too sure.”. Alex remembers their bracelet, and is calmed by the presence of their friend. If something would happen Alex could send their coordinates and an alarm. The other hiker slowly approaches and the anxiety grows. But then the other hiker just passes, with a simple “Good day”, and the release of tension in the chest is immediate. “Nothing happened. Phew. But I’m still glad my friend was with me.”

Semi-structured interviews were used in order to let the participants think and talk freely while still staying on topic and allowing the interviewer to ask follow-up questions and probe further[9]. Additionally, open-ended questions were used since the primary objective was to collect a wide range of data and insights given the early stage of the project. Probing was used to get an in-depth understanding of user perspectives. A group interview was conducted in order to facilitate discussion amongst the participants.

4.2 User feedback

Whilst answering questions regarding hiking alone all participants expressed discomfort with the idea. One woman said that “Nice when no-one is here but also ‘oh no, no-one is here’”. All participants said that they thought the bracelet was a good idea as most respondents would not hike alone due to safety concerns. A common concern was being in a remote location where help might not be easily accessible, especially at night or during winter months.

Most stated that they would want to easily be able to contact someone or at least have a friend or family be aware of their location to feel safer. “The idea that someone, even if it’s just one person, knows where I am gives me comfort. It’s nice that someone could alert the police.”. Some participants also mentioned that having parents know they are safe would be reassuring, especially for those far from home. Relating to that the ability to send different levels of alerts (emergency vs. check-in) was valued. Some even meant that they would want something in between, being able to send their partner bracelet a heads-up that a dangerous situation might arise.

Regular check-ins were seen as useful for ensuring safety over long hikes, with the system alerting the friend if the hiker failed to respond within a set period. GPS coordinates were considered an essential feature for locating the hiker quickly in an emergency. Participants expressed a preference for both visual (blinking lights), vibration, and in some cases, even audible alerts as well. But they all agreed that the most important feature was a simple and efficient alert system that could be activated in moments of panic. Many users emphasized that it should be easy to trigger, such as holding down a button or using voice activation rather than pressing multiple times and should avoid accidental activations. One user said “yeah a button, i think a lot of our senses get unconnected if you get attacked, so it should be easy.”

When asked about extra features like crash detection, to be able to track and react to vital signs, like heart rate and oxygen, opinions varied. Some felt these features could lead to false alarms, while others found them useful in certain emergency scenarios. There were hesitations about contacting authorities directly, all female

participants preferred to contact their friend firsthand as if to not be scared to use the feature or to activate it by mistake. The male participant wished for the distress signal to be sent to authorities directly.

The bracelet's aesthetics was also an important topic with the interviewees, with comfort and it not being irritating to wear being the most important. They wanted something durable, that wasn't ugly and even be customizable. Silicone materials were less favored, with a preference for something that feels good on the skin, and that breathes. Many said they would wear the bracelet more often if it looked appealing, even daily. But some were cautious about wearing it daily due to fears of triggering constant anxiety in them. They also questioned it being two different bracelets, and asked if it could maybe be combined into one so the user would not have to purchase two different bracelets.

They did not have as many comments on the friends bracelet but they liked that if they had a friend or family member hiking alone in remote places that they would get updates that they were ok and be alerted if not. They preferred a combination of vibration, sound, and GPS tracking. They expressed a strong desire to receive alerts directly on the bracelet, rather than through a phone, for a more seamless and immediate response.

5 FINAL PROTOTYPE

5.1 Design changes and refinement

Based on the user feedback received during the pilot evaluations several design decisions were made. The two bracelets were merged into a single unit, streamlining the design process by reducing both complexity and responding to the user's need of not having to purchase multiple bracelets. It is important with unambiguous feedback and a straightforward interface when designing for stressful situations [9]. Additionally, the emergency tag for extreme hikers was removed, better suited for recreational hikers, minimizing accidental triggers and in favor of condensing the design for ease of handling. A warning mode was also included in the design based on the wish of having something between a check-in and emergency distress.

Furthermore, the feedback system was refined to include auditory, haptic and visual signals in order to not exclude people with disabilities and to offer a redundancy in information transfer. The GPS coordinates are now sent as a SMS text-message for the same reason, since the phone has built in functions to read texts out loud for the visually impaired. Moreover, the redundancy of the SMS and the bracelets distress-mode will ensure that whoever is on the receiving end gets the alert, even if they would not have their phone or the bracelet nearby.

For the final prototype the first step was to prepare the electronics. *Arduino Uno* was used for that, it had a button connected to it for the alerts and a LED strip to showcase the different color modes; white, green, yellow and red (see Figure 6, 7). Using Arduino to simulate the feedback system was an effective way to explore the solution and iteratively come up with the optimal use for the color modes. For instance, the green was exchanged to a white color to signal the ideal state, so that the green color could be used to express confirmation.

The bracelet band itself was hand sewn from a comfortable fabric and made large enough to fit the LED strip and button and the cords connecting them to the Arduino within. It was also broad enough so that the 3D printed links could be attached through sewing. Finally, the plastic links to cover the LED strip were created using *Alias Fusion 360* and 3D printed using transparent PLA filament. The links were designed in a manner so they could click together and could rotate independently from each other, which was necessary in order for the bracelet to wrap around the wrist. The links had a groove underneath where the LED strip and cords for the button could fit and on the end a link was made with a hole for the button. Once all of those individual components were ready the assembly of the bracelet was done (see Figure 3).

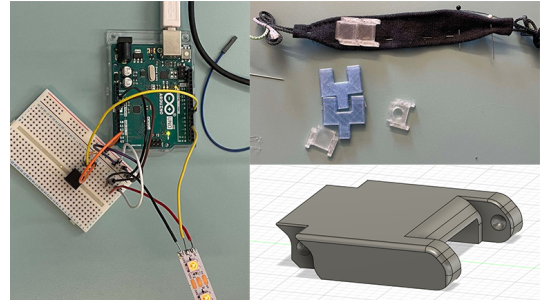


Figure 3: Photo of the prototype in progress, audrino, sewing, and CAD

5.2 TrailMate

TrailMate is a smart security bracelet designed for hikers, to help them feel safer while in more remote areas. The system consists of two bracelets: one worn by the hiker and the other given to a trusted partner. Both bracelets are identical, each equipped with two modes toggled with a switch: "home" and "hike". The system communicates different alarm levels through visual, haptic and auditorial feedback. See the table below for interaction and corresponding feedback between the hiker and person at home (see Figure 4).

Alarm level	Hike-mode interaction	Hike-mode feedback	Home-mode interaction	Home-mode feedback
All good	No interaction	White	No interaction	White
Confirmation	No interaction	Green to white	No interaction	Green to white
Self check-in	Press button within 1 min	Vibration & yellow	No interaction	White
Friend check-in	Press button within 1 min	Vibration & yellow	Press button	Yellow until answered
Warning	Switch	Vibration & blink yellow	Receives text	Yellow until changed
Distress signal	Hold down button for 3s	Vibration, red & sound	Receives text	Vibration, red & sound
Turn off distress	Switch	Green to white	No interaction	White

Figure 4: Table describing the bracelets' different functionalities

The partner using the home mode can initiate a check-in by pressing a button on their bracelet, causing it to turn yellow while awaiting a response (see Figure 5c). Simultaneously, the hiker's

bracelet also turns yellow and vibrates to signal the incoming check-in. Once the hiker responds by pushing the button on their bracelet, both bracelets flash green (see Figure 5a), confirming the communication before returning to their idle state, with white colored lights (see Figure 5b). If the hiker fails to respond within one minute, the partner's bracelet triggers an alert by turning red, vibrating, and sending the hiker's GPS coordinates as an SMS the partner's phone, allowing immediate action to be taken (see Figure 5d). Additionally, the hiker can schedule regular check-ins at set intervals, such as every couple of hours. If the hiker misses a response, the system alerts the partner in the same manner. In situations where the hiker feels endangered, they can manually trigger the alert by holding the button for three seconds, sending an immediate distress signal to the partner's bracelet (see Figure 5d). If the hiker feels like a dangerous situation might arise, they can send a warning to the partner bracelet, causing the partner bracelet to turn yellow, and the hikers to vibrate once and blink yellow until further action is taken. The warning- and distress mode can both be manually turned off with a switch, if they were accidentally triggered or the danger has passed.

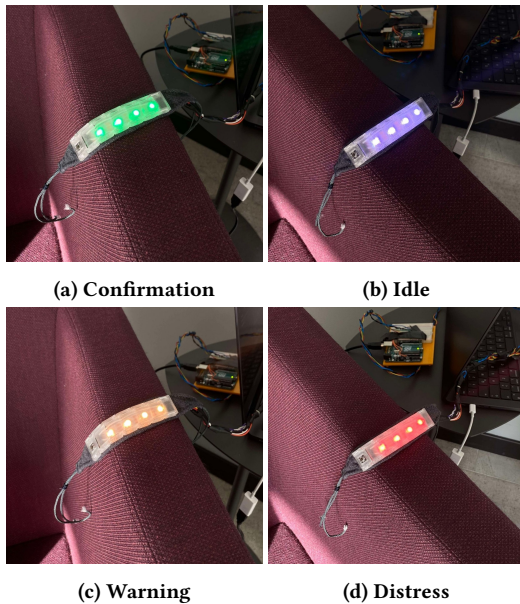


Figure 5: Different colors of the visual feedback system

Due to time constraints, certain features such as the warning mode, “turn-off distress” function, home/hike mode switch, and on/off button were not included in the physical prototype. However, these features are still functions in the final idea.

6 EVALUATION

The last prototype was evaluated through two semi-structured group interviews, each with two students aged 20-30. The first group consisted of two women, while the second group included one woman and one man.

When asked how the bracelet would make them feel while hiking, the first group expressed that it would bring them a sense of



Figure 6: Hikers bracelet in hike-mode



Figure 7: Partner bracelet in home-mode

reassurance, stressing that one can not be sure to always have data on their phone. They added that the device could also be useful when walking alone at night, for example when walking home from a night out. Additionally, they highlighted that they liked the function of the red light since it might scare off a potential perpetrator. The other group, however, expressed concern about the red light being so obvious. They argued that the perpetrators' awareness of the bracelet might cause them to either rip it off or send a false reassurance that the hiker is okay. They were also concerned the red light might cause stress to the perpetrator making them more dangerous. Overall they thought that although it would not protect them from immediate danger it would make them feel safer to have it than not.

The groups were also asked about how they would feel wearing the bracelet in home-mode while their partner was hiking alone. The first group said that it would be a comforting reassurance knowing if their partner was safe or not. One participant explained that “I would feel okay, she is safe and if she is not, I will know.” Still they expressed that they would be worried in case their partner would get injured and not be able to push the button. They did however recognize that the self-check-in might help in that scenario. One participant said “I would love to be at the other end of this”. The woman in the second group did not find the partner bracelet necessary, instead, she would rather use her phone for the features of the partner's bracelet. In contrast to her opinion, the man appreciated that the person at home had their own bracelet. He preferred being able to distance himself from the phone since it might distract him while, for example, working.

When asked about improvements, both groups agreed the bracelet could be more delicate. The second group again highlighted how a more discrete wristband could be better since it won't catch the

perpetrators' attention. They also thought the appearance could be improved by dimming the lights or replacing them with vibrations. A woman in the second group proposed an alternative solution, making it a necklace. She argued that people naturally tend to fidget with necklaces, which would make pressing a button less noticeable. She said that "You could play with it and it would not be obvious that you are sending a distress signal.". The other group however said that a necklace would be a bad idea since the wearer wouldn't see the alerts. They suggested an alternative design for the device, a ring, allowing the button to be pressed with one hand in situations where the other hand might be stuck, injured, or restrained. They also discussed technical aspects such as battery life and indication for remaining battery.

Whether they could see themselves using the device or not, the first group recognized the bracelet's value but were skeptical whether they would use it themselves. They were more inclined toward its purpose after a night out, when walking home alone. In the second group, one interviewee said they would use the bracelet for hiking, while the other said they would mostly use it on dangerous trails.

Furthermore, both groups were asked to describe the product with one word. Some of the words used to describe it were: effective, comforting, direct, connected, safety, responsibility and 'peace of mind'.

7 DISCUSSION

Many stressful situations can occur when hiking alone. Through the evaluations we could determine that TrailMate is a desired product, where the participants expressed that it would make them feel "safe" and "calm". Sicorello et al. proved that social interactions can lower stress levels in something called stress buffering [10]. This can, interestingly, be achieved by solely indicating the presence of social interactions. Moreover, they also found that stressful situations create a need for social company. Through the use of TrailMate one could not only have lowered levels of stress hormones but possibly feel encouraged to take action in stressful situations. Since there are few and simple directions of action that are directly available to the user, the need for decision making is minimized. What's interesting is the fear of directly contacting the authorities that became clear during the pilot tests. Having a response directly connected to local authorities seemed to trigger self-doubt. Whereas having the bracelet be connected to a trusted partner seemed to encourage use, which in itself might be a critical factor in stressful situations where quick decisions need to be made.

Moreover, when considering ethics it is important to discuss how a product like this creates a false sense of security where the trust for the product encourages reckless behavior. In the possibility that the attacker disarms the hiker wearing the bracelet and somehow turns it off or destroys it – would the person wearing the partner bracelet react with less urgency if the hiker didn't come back on time since they "have the bracelet and it would have alerted them otherwise"? More importantly, do these factors outweigh the need for the bracelet?

With regard to the methods used some approaches might have affected the outcome of the project. One reason being that we went directly from evaluating a low-fidelity prototype to constructing

a rather high-fidelity one. It would have given useful data to host evaluations during the development, either with some mid-fidelity prototype or with parts of the prototype, such as the feedback system. In the final prototype several design decisions that had been made would now be cost and time inefficient to change. Evaluating a mid-fidelity prototype would have added further iterations to the process, ensuring more reliable results [9]. Moreover, when choosing the winning concept during ideation a more scientific approach could have been applied than the one used. One approach could have been to more carefully define the user needs and requirements and evaluate the concepts using a Pugh-matrix in order to not have the team's individual biases influence the decision [8].

During the evaluations semi-structured interviews were used. However, in the last evaluation it might have been more useful to use structured interviews. This would have gathered more focused data [9]. However, one can argue that our final prototype is still in the beginning of development, and therefore using structured interviews might have limited us rather than helping, since it would call for clearly understood goals. The final interviews were still exploratory and gathered data necessary for further development, and therefore using semi-structured interviews was a beneficial approach. Even so, the sample size was small, and to ensure representative data more participants from a broader background would have been needed [9].

There was doubt expressed during the final evaluation about whether the bracelets' distress function should be bright, red, and loud. Two of the participants argued that this might aggravate a potential attacker whilst the other two participants claimed the opposite, that it might scare away a potential attacker. We know that for example burglar alarms are loud in order to scare away the intruder, but what is true if the hiker and the attacker are alone in a remote location? Will the notion that someone has been alerted act as a warning or a threat? Regardless, the thought that the warning might aggravate the attacker might be enough to discourage the hiker from alerting their partner and should be considered and evaluated further during future work.

8 CONCLUSION

This study explores how a wearable device could benefit solo hikers by fostering a sense of companionship, enhancing safety, and supporting emotional well-being. Findings from related work indicates that devices that improve safety for women and other minority groups are needed. Our evaluations also confirm the value of having a back-up safety net when hiking alone. The process of iterative prototyping has been proven to be efficient, where user feedback effectively drove the development towards a user-centered product. Final evaluations revealed that the bracelet enhances the sense of safety through the presence of social interaction when engaging in hiking. Additionally, it provides reassurance to those close to the hiker. However, to make sure the bracelet fulfills its purpose further development is required. The aesthetics of the bracelet needs to be improved, and the functions need to be further examined to achieve its full potential.

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